ETH zürich

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Spiking Neural Networks for Ultra-Low Power Event-Based Optical Flow Estimation

Integrated Systems Laboratory (ETH Zürich)

candidate: Lorenzo Squarzoni

advisor: Prof. Dr. Francesco Conti
supervisors: Dr. Lorenzo Lamberti
Dr. Daniele Palossi
Dr. Francesco Barchi
Prof. Dr. Andrea Acquaviva
Prof. Dr. Luca Benini

PULP Platform
Open Source Hardware, the way it should be!

pulp-platform.org
@pulp_platform
company/pulp-platform
youtube.com/pulp_platform

Context and Motivation

Autonomous nano-sized drones


environmental monitoring


surveillance

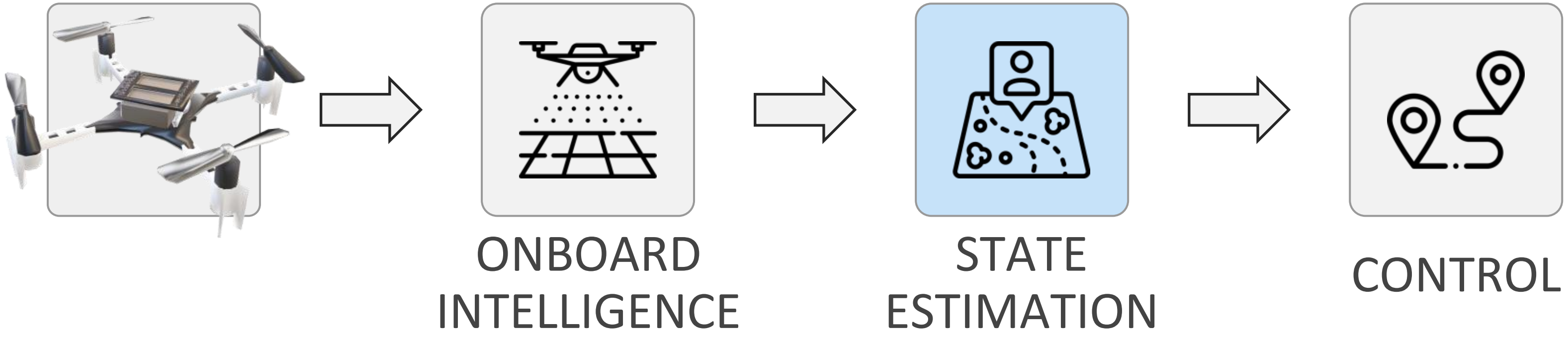
30g
10cm
~30MB
~100mW


industrial inspection


search and rescue missions

Context and Motivation

Enabling real-time autonomous navigation

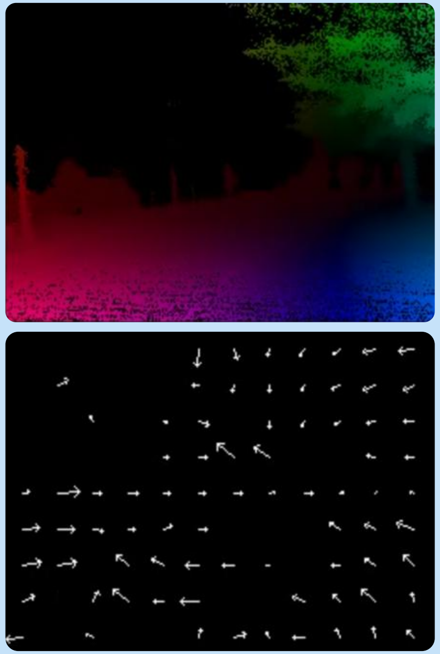


Context and Motivation

Enabling autonomous navigation

ONBOARD INTELLIGENCE

OPTICAL FLOW ESTIMATION
to measure pixel-wise motion from vision



CONTROL

Context and Motivation

Enabling autonomous navigation

ONBOARD INTELLIGENCE

EVENT-BASED CAMERAS

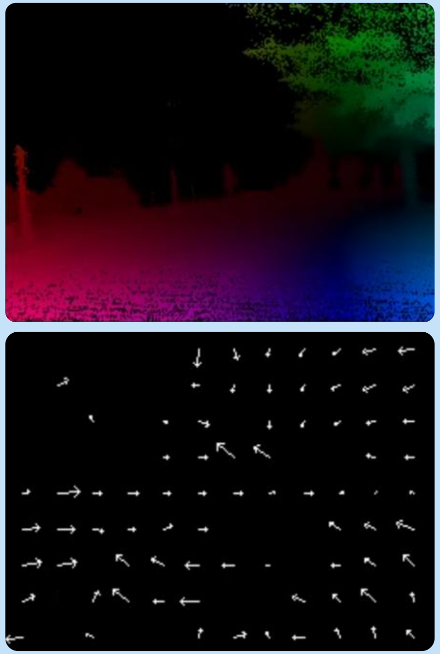
high temporal res. [~µsec]

high dyn. range [~140dB]

low power

OPTICAL FLOW ESTIMATION

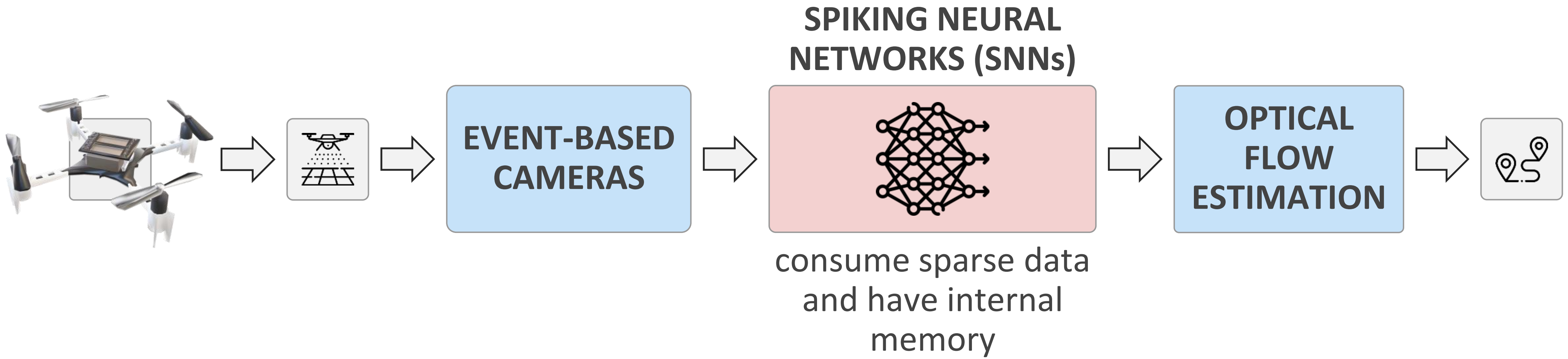
to measure pixel-wise motion from vision



CONTROL

Context and Motivation

Enabling autonomous navigation



Context and Motivation

Enabling autonomous navigation

low power consumption,
effective for low inference latency

EVENT-BASED CAMERAS → SPIKING NEURAL NETWORKS (SNNs) → OPTICAL FLOW ESTIMATION

LIF neurons have memory, providing
temporal correlation

Related works and challenges

State of the art: SNNs for event-based optical flow [1]

what have been explored?

ARCHITECTURE

fully spiking
models

[1] **Federico Paredes-Vallés et al.**, "Self-Supervised Learning of Event-Based Optical Flow with Spiking Neural Networks", NeurIPS 2021

Related works and challenges

State of the art: SNNs for event-based optical flow [1]

what have been explored?

ARCHITECTURE	TRAINING
fully spiking models	**stable** training (surrogate gradient)
	effective motion compensation **loss**

ETH zürich ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

[1] **Federico Paredes-Vallés et al.**, "Self-Supervised Learning of Event-Based Optical Flow with Spiking Neural Networks", NeurIPS 2021

Related works and challenges

State of the art: SNNs for event-based optical flow [1]

what have been explored?

ARCHITECTURE	TRAINING	PERFORMANCE
fully spiking models	**stable** training (surrogate gradient) effective motion compensation **loss**	**accuracy** comparable to Artificial Neural Network (ANN) models

[1] **Federico Paredes-Vallés et al.**, "Self-Supervised Learning of Event-Based Optical Flow with Spiking Neural Networks", NeurIPS 2021

Related works and challenges

State of the art: SNNs for event-based optical flow [1]

what have been explored?

ARCHITECTURE → **TRAINING** → **PERFORMANCE** → no on-board **deployment** on nano-drones

fully spiking models

stable training (surrogate gradient)

effective motion compensation **loss**

accuracy comparable to Artificial Neural Network (ANN) models

GPU
~600W
>24GB

ULP MCU
~100mW
~30MB

[1] **Federico Paredes-Vallés et al.**, “Self-Supervised Learning of Event-Based Optical Flow with Spiking Neural Networks”, NeurIPS 2021

Related works and challenges

State of the art: SNNs for event-based optical flow [1]

what have been explored?

this thesis:

ARCHITECTURE → **TRAINING** → **PERFORMANCE** → **ULP MCU DEPLOYMENT**

fully spiking models

stable training (surrogate gradient)

effective motion compensation **loss**

accuracy comparable to Artificial Neural Network (ANN) models

ULP MCU
~100mW
~30MB

ETHzürich — ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

[1] **Federico Paredes-Vallés et al.**, "Self-Supervised Learning of Event-Based Optical Flow with Spiking Neural Networks", NeurIPS 2021

Contributions

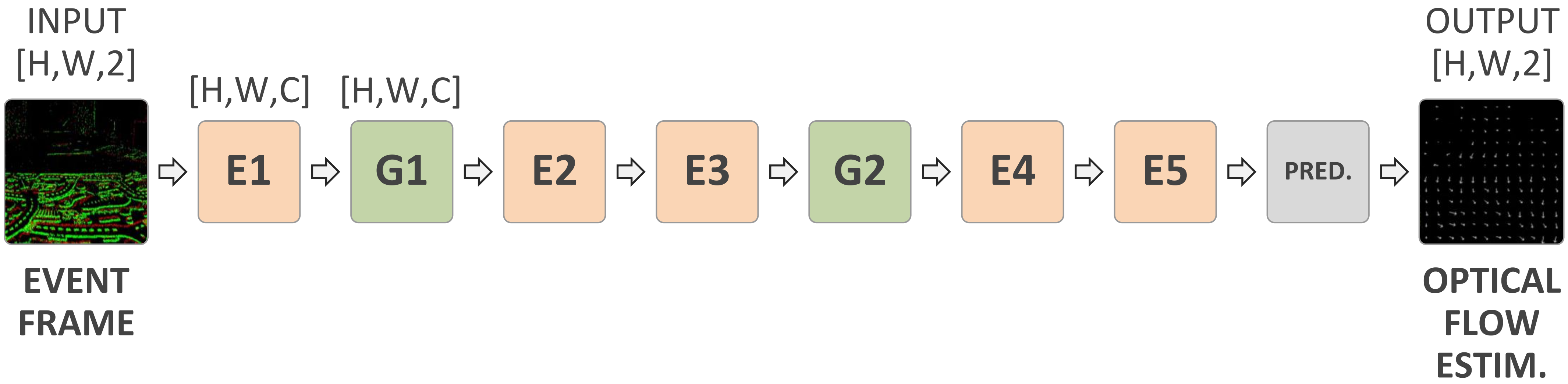
What this thesis delivers:

Hardware-aware **shrinking** and **quantization** of a state-of-the-art SNN optical flow model

Deployment on a PULP microcontroller, with measurements of memory, latency and throughput

Background

Software: spiking FireNet for optical flow

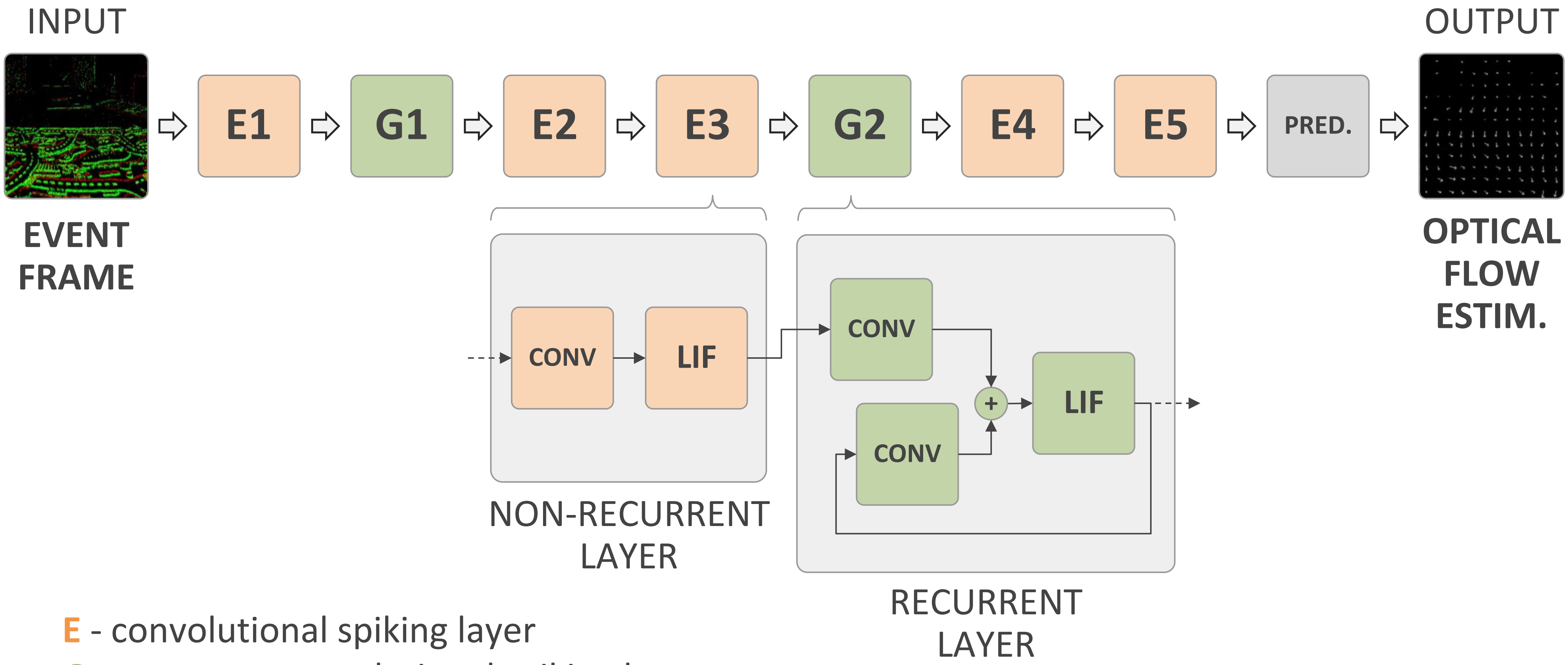


E - convolutional spiking layer

G - recurrent convolutional spiking layer

Background

Software: spiking FireNet for optical flow

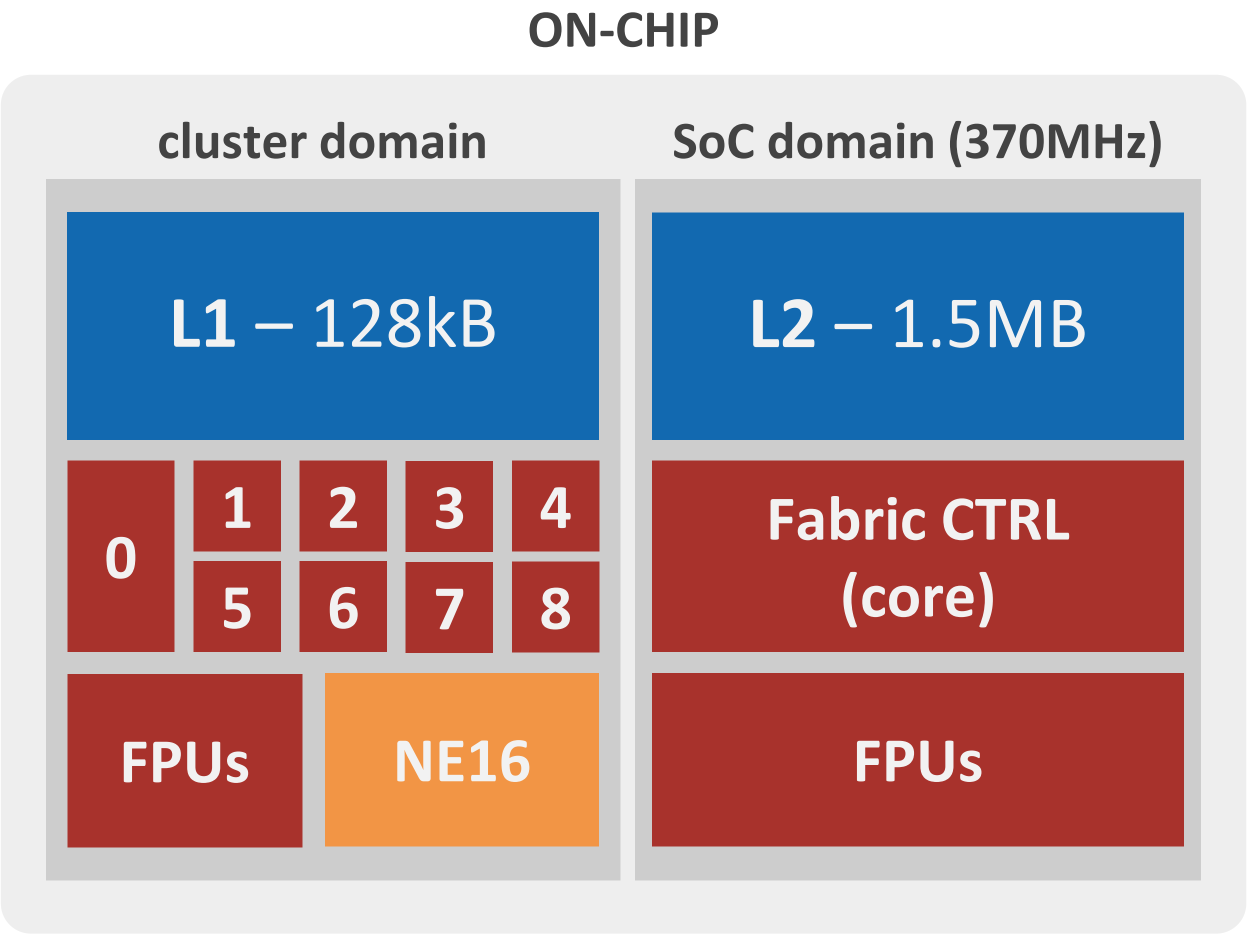


E - convolutional spiking layer

G - recurrent convolutional spiking layer

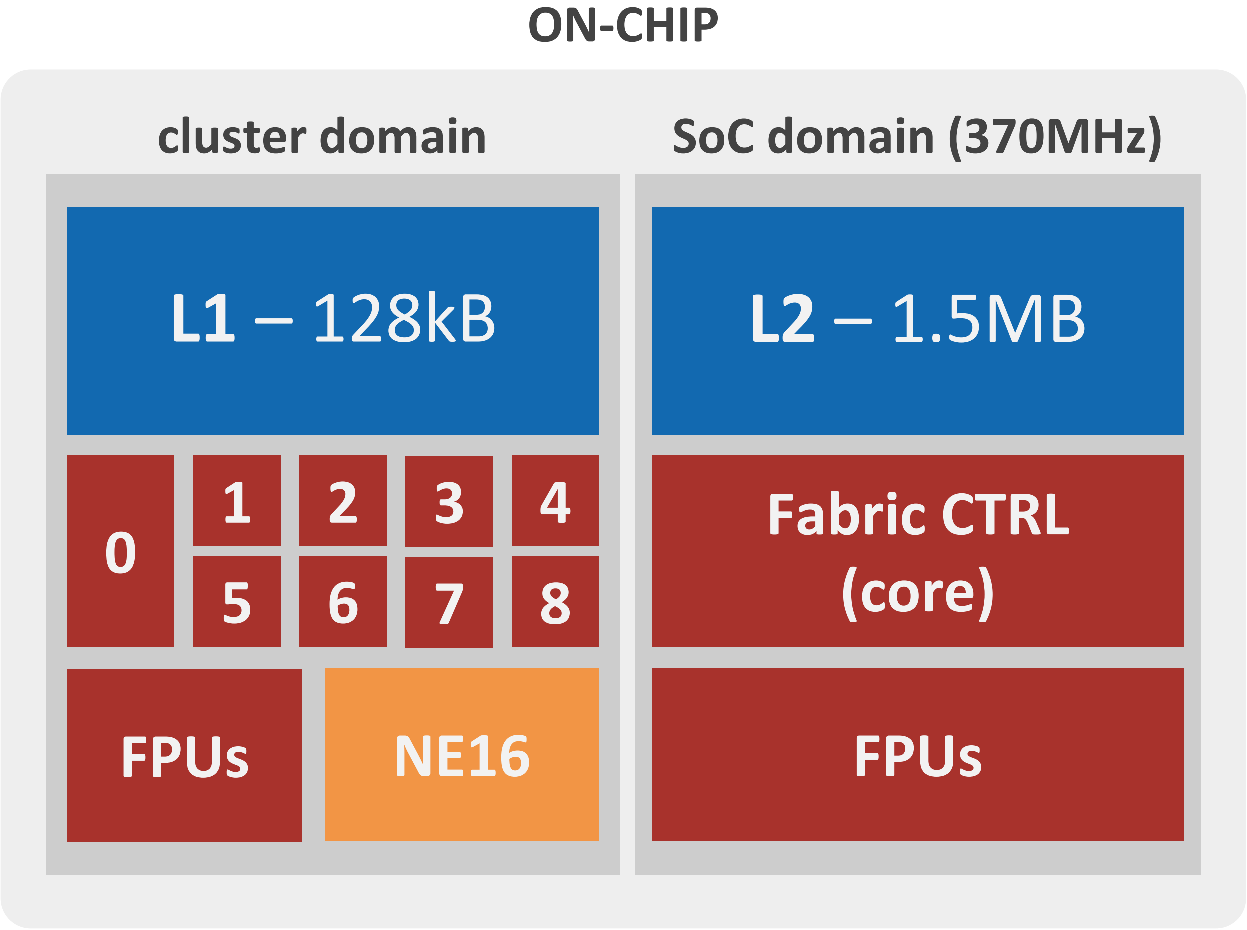
Background

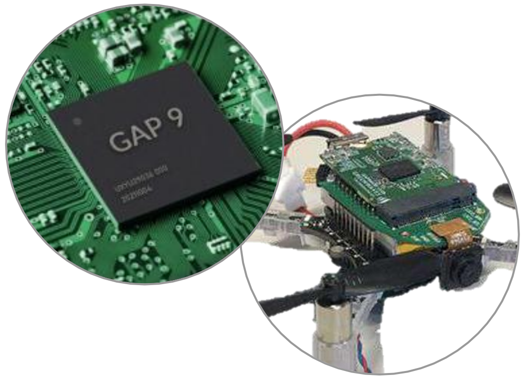
Target hardware: PULP GAP9 + GAP9 shield

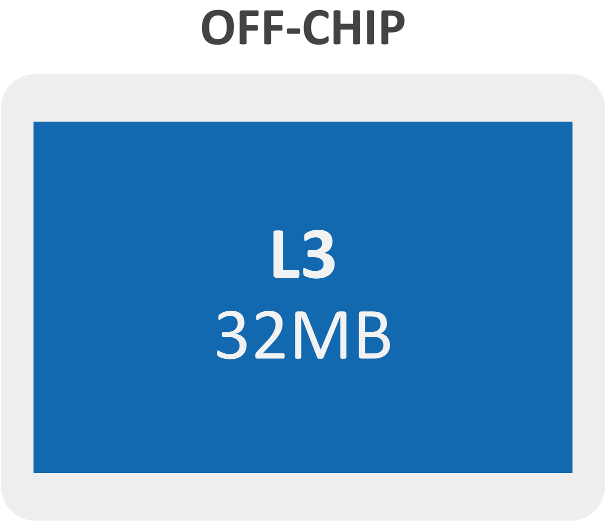


Background

Target hardware: PULP GAP9 + GAP9 shield

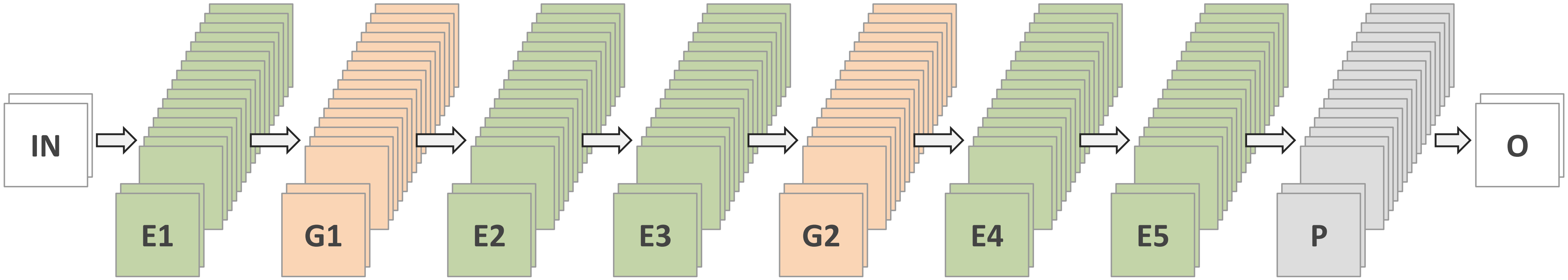






Methodology

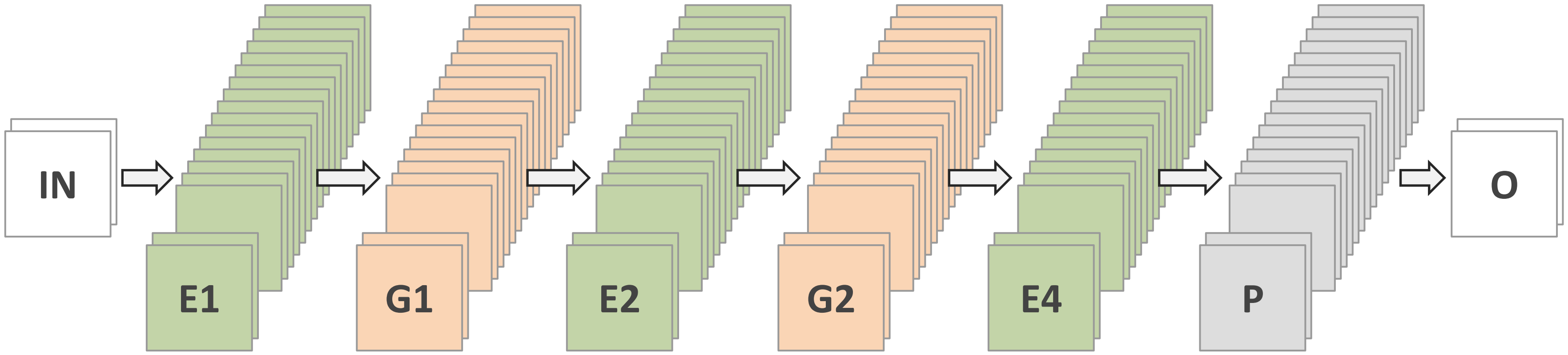
Architecture exploration: shrinking and pruning



Methodology

Architecture exploration: shrinking and pruning

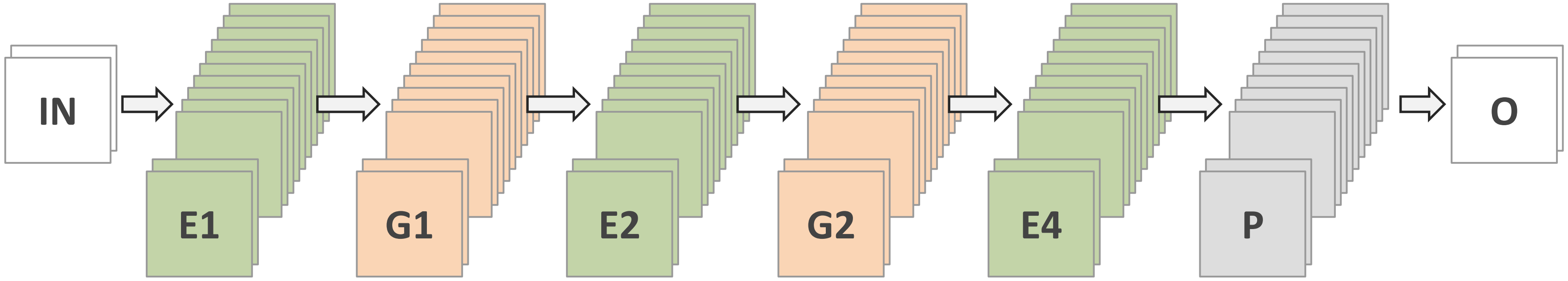
- non-recurrent **LAYERS REMOVAL**



Methodology

Architecture exploration: shrinking and pruning

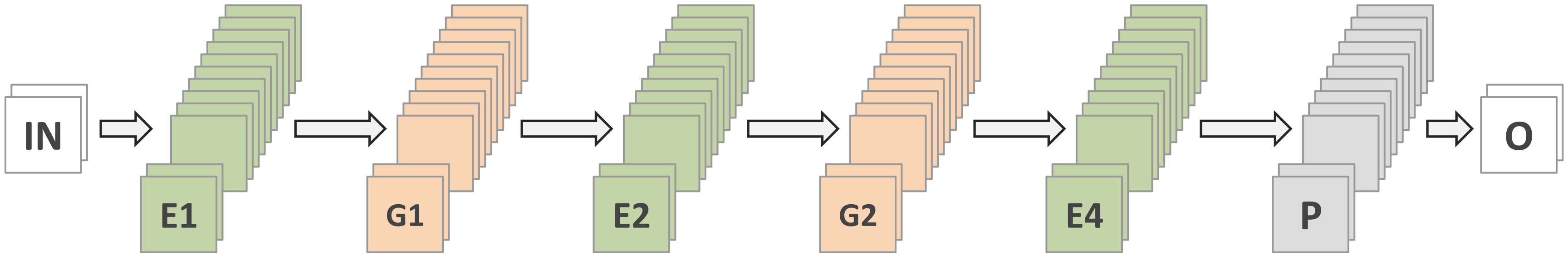
- non-recurrent **LAYERS REMOVAL**
- **REDUCTION** in the **N. of CHANNELS**



Methodology

Architecture exploration: shrinking and pruning

- non-recurrent **LAYERS REMOVAL**
- **REDUCTION** in the **N. of CHANNELS**
- **INPUT SPATIAL RESOLUTION REDUCTION** by average pooling



Methodology

Training to hardware pipeline

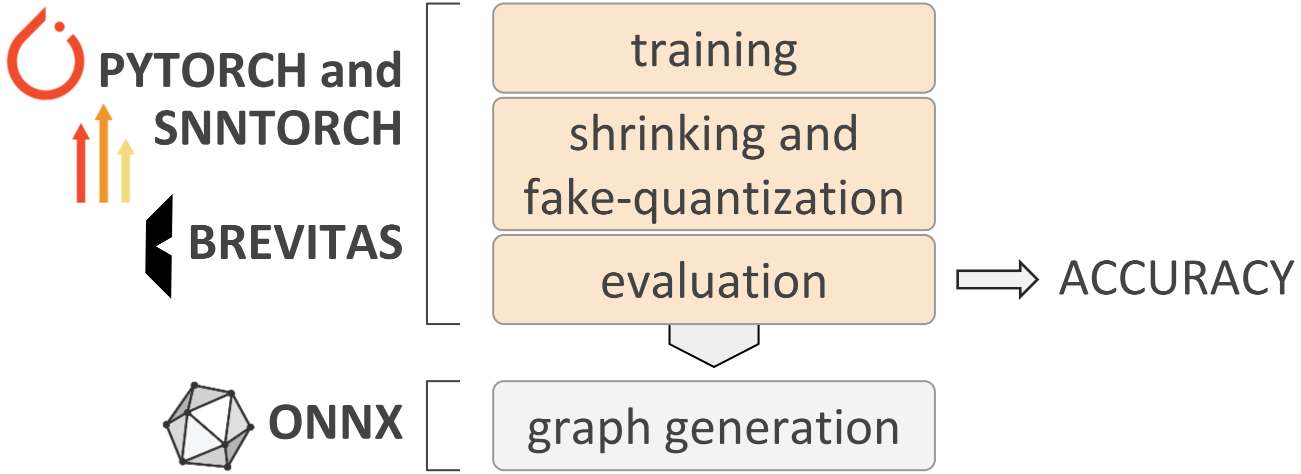
PYTORCH and SNNTORCH

BREVITAS

- training
- shrinking and fake-quantization
- evaluation ⟹ ACCURACY

Methodology

Training to hardware pipeline



Methodology

Training to hardware pipeline

PYTORCH and SNNTORCH
BREVITAS

- training
- shrinking and fake-quantization
- evaluation ⇒ ACCURACY

ONNX

- graph generation

DEEPQUANT

- real quantization

Methodology

Training to hardware pipeline

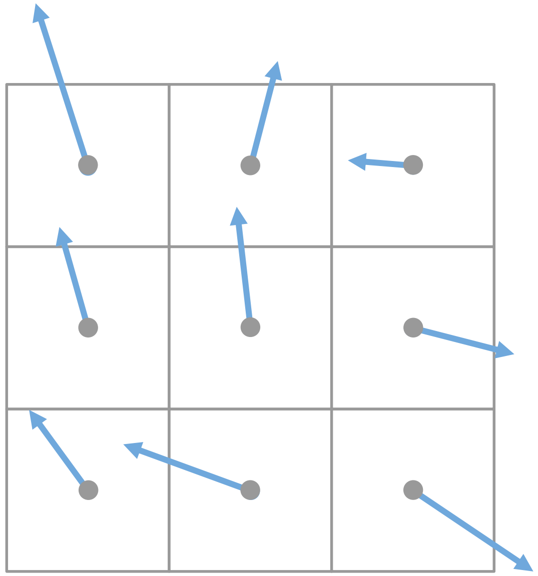
PYTORCH and SNNTORCH, BREVITAS	training
	shrinking and fake-quantization
	evaluation ⇒ ACCURACY
ONNX	graph generation
DEEPQUANT	real quantization
DEEPLOY [1] and PULP-NN [2]	C code generation and optimization
GVSOC [3]	simulation ⇒ MEMORY ALLOCATION, LATENCY, THROUGHPUT

[1] Deeploy: Enabling Energy-Efficient Deployment of Small Language Models On Heterogeneous Microcontrollers, arXiv:2408.04413
[2] PULP-NN: Accelerating Quantized Neural Networks on Parallel Ultra-Low-Power RISC-V Processors, arXiv:1908.11263
[3] GVSoC: A Highly Configurable, Fast and Accurate Full-Platform Simulator for RISC-V based IoT Processors, arXiv:2201.08166

Results

Metric: Average Angular Error (AAE)

E - estimated flow vectors



PIXELS

ERROR COMPUTATION

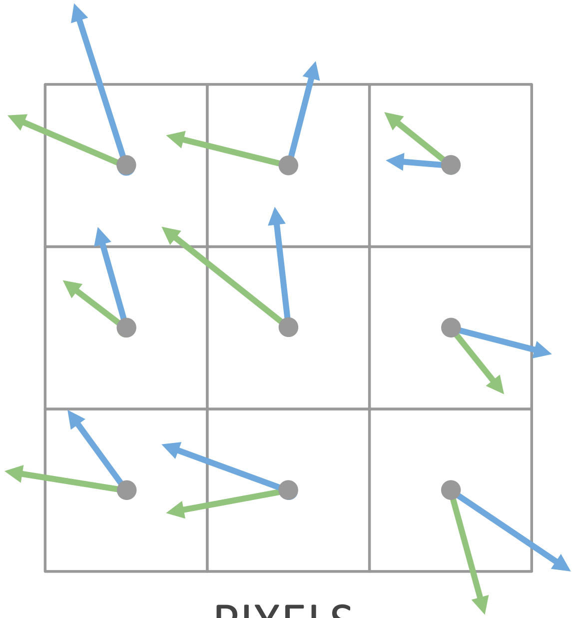
$$AAE = \frac{\sum_{\text{pixels}}\left(\cos^{-1}\frac{\overline{GT}\cdot\bar{E}}{\|\overline{GT}\|\cdot\|\bar{E}\|}\right)}{\text{N. pixels with events}}$$

Results

Metric: Average Angular Error (AAE)

E - estimated flow vectors

GT - ground-truth flow vectors



PIXELS

ERROR COMPUTATION

$$\text{AAE} = \frac{\sum_{\text{pixels}}\left(\cos^{-1}\frac{\overline{\mathbf{GT}}\cdot\bar{\mathbf{E}}}{\|\overline{\mathbf{GT}}\|\cdot\|\bar{\mathbf{E}}\|}\right)}{\text{N. pixels with events}}$$

ETH zürich

ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

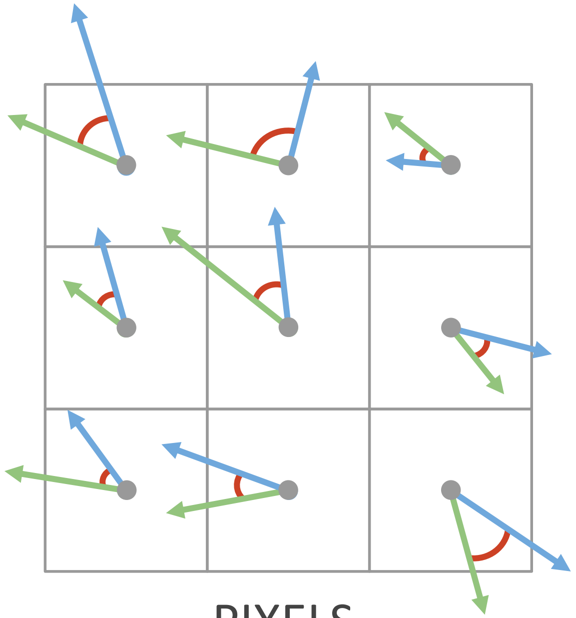
Results

Metric: Average Angular Error (AAE)

E - estimated flow vectors

GT - ground-truth flow vectors

AE - angular error



ERROR COMPUTATION

$$\mathbf{AAE} = \frac{\sum_{\text{pixels}}\left(\cos^{-1}\frac{\overline{\mathbf{GT}}\cdot\bar{\mathbf{E}}}{\|\overline{\mathbf{GT}}\|\cdot\|\bar{\mathbf{E}}\|}\right)}{\text{N. pixels with events}}$$

Results

Baseline model: spiking FireNet [fp32]

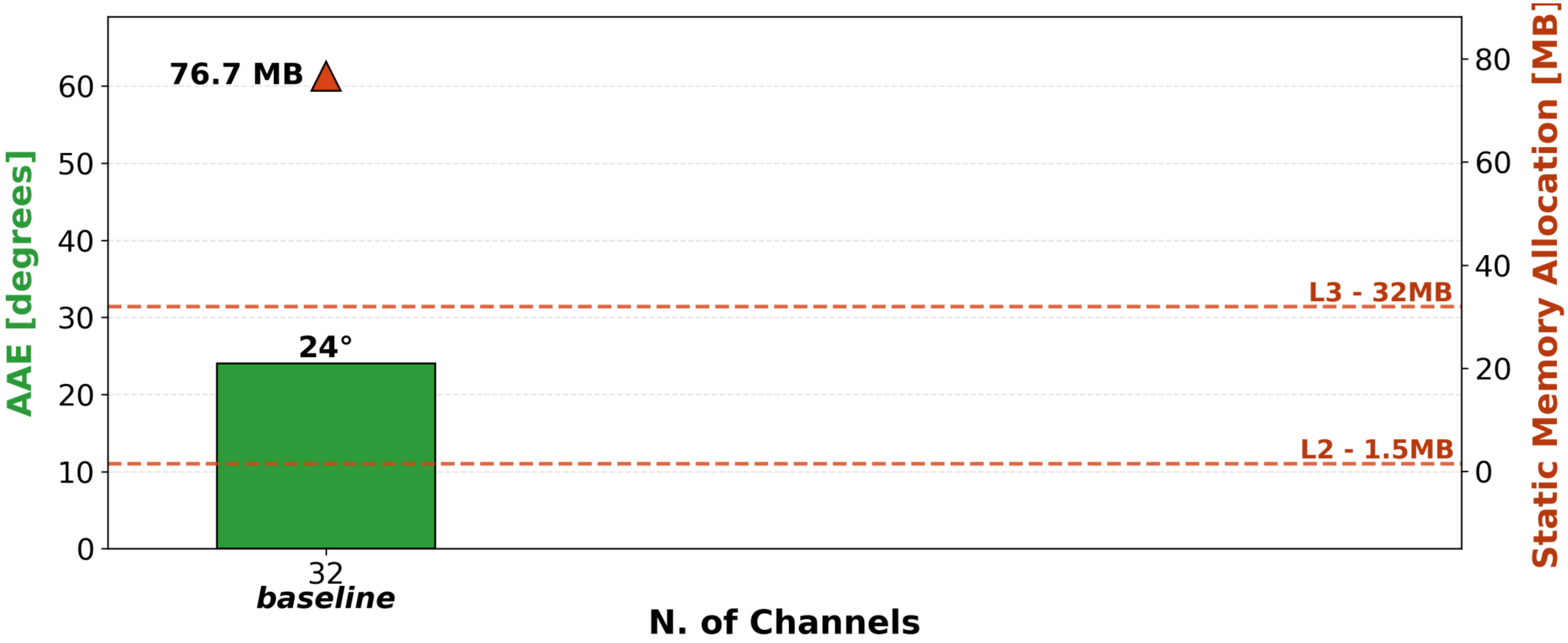
32 channels, 8 layers and **256x256** pixels of resolution

AAE ↓ [degrees]	Static Memory Allocation [MB]	N. operations [G MACs]
24°	**76.7**	**7.35**

good angular accuracy,
but doesn’t fit in L3

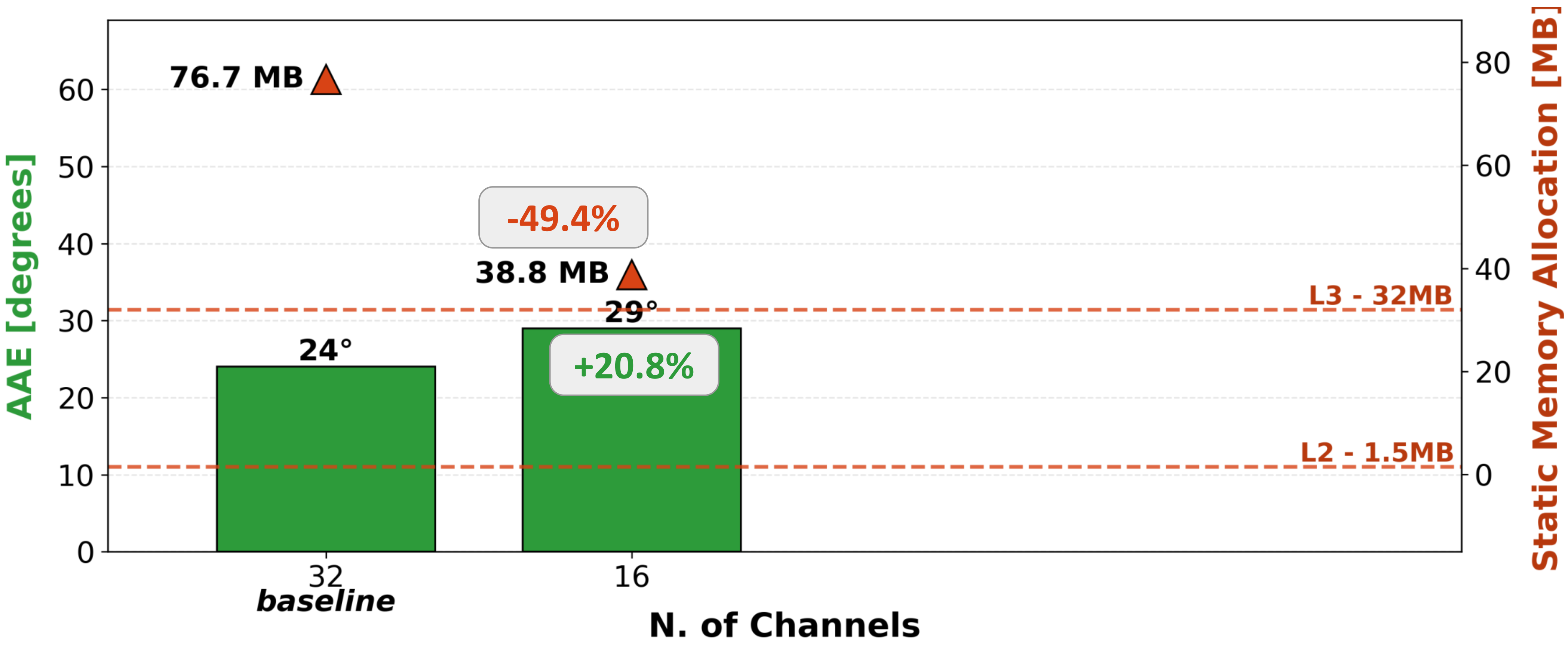
Results

Analysis on: N. of CHANNELS



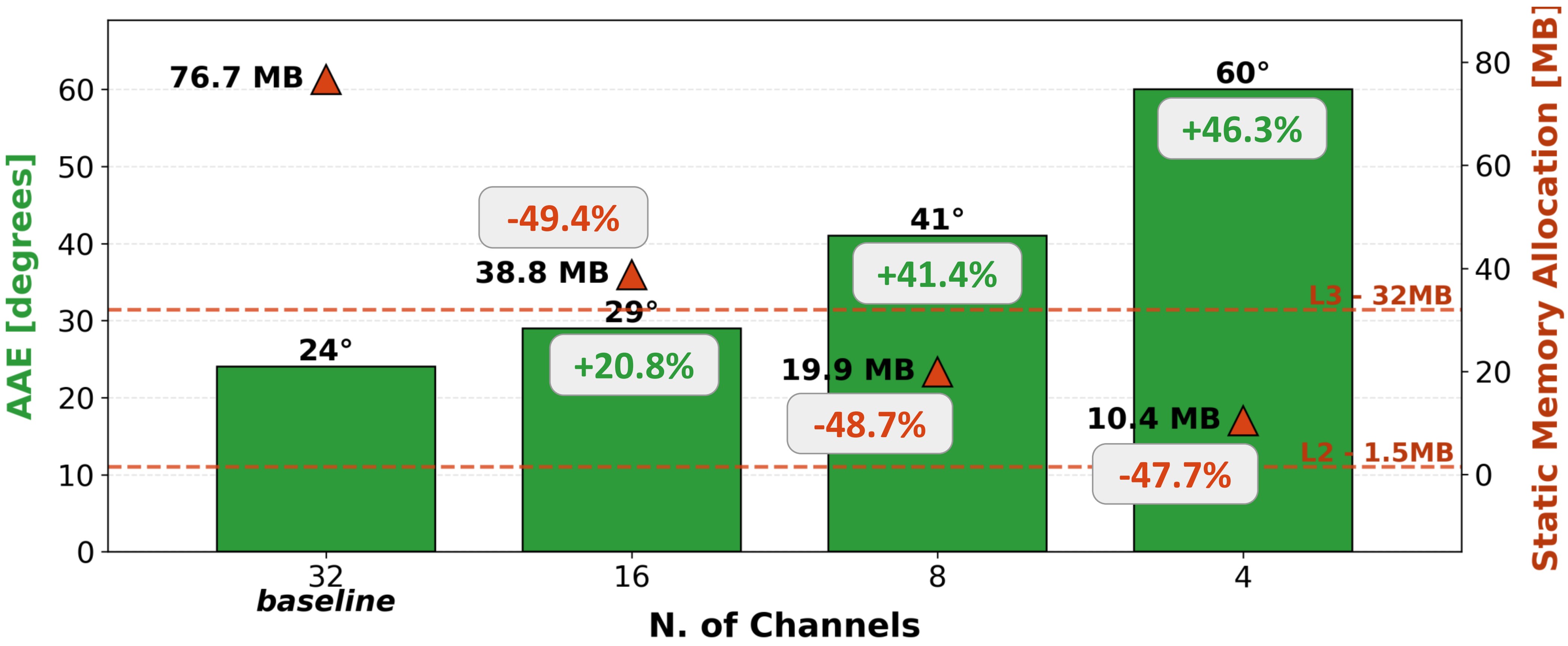
Results

Analysis on: N. of CHANNELS



Results

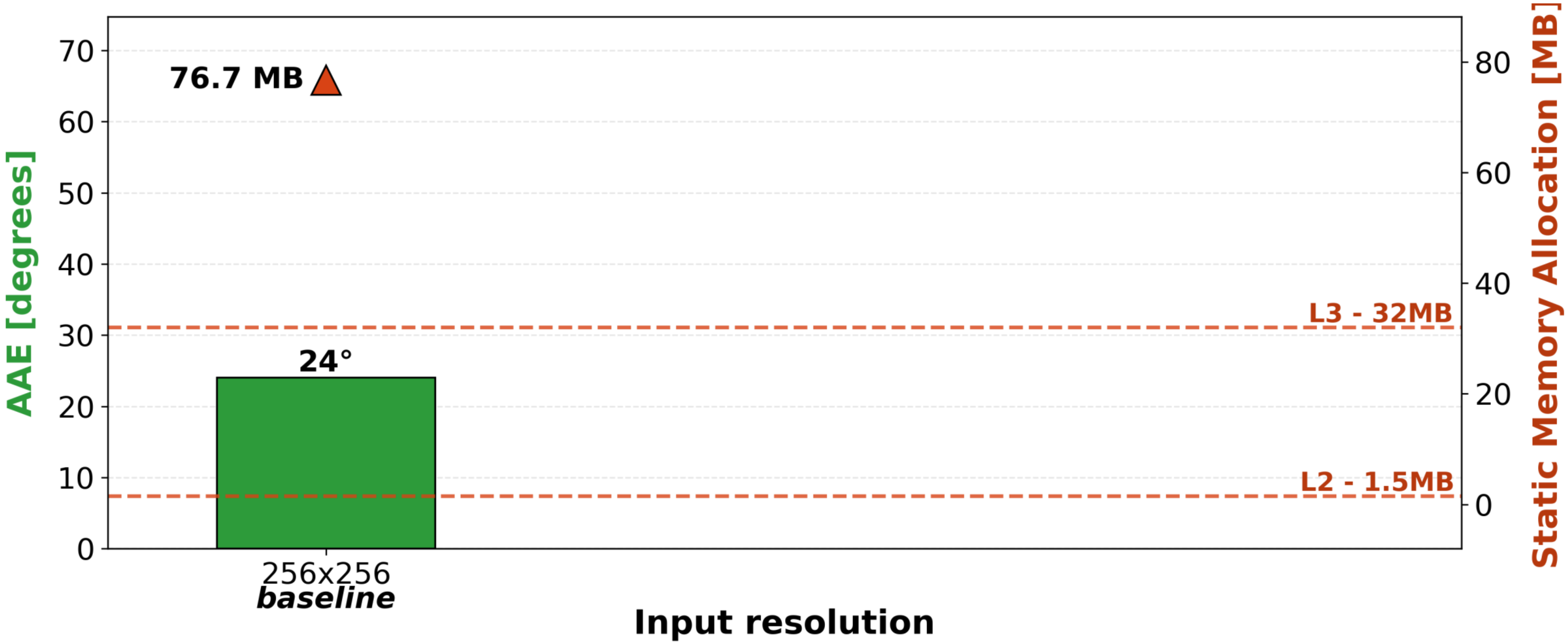
Analysis on: N. of CHANNELS



ETH zürich ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

Results

Analysis on: INPUT RESOLUTION



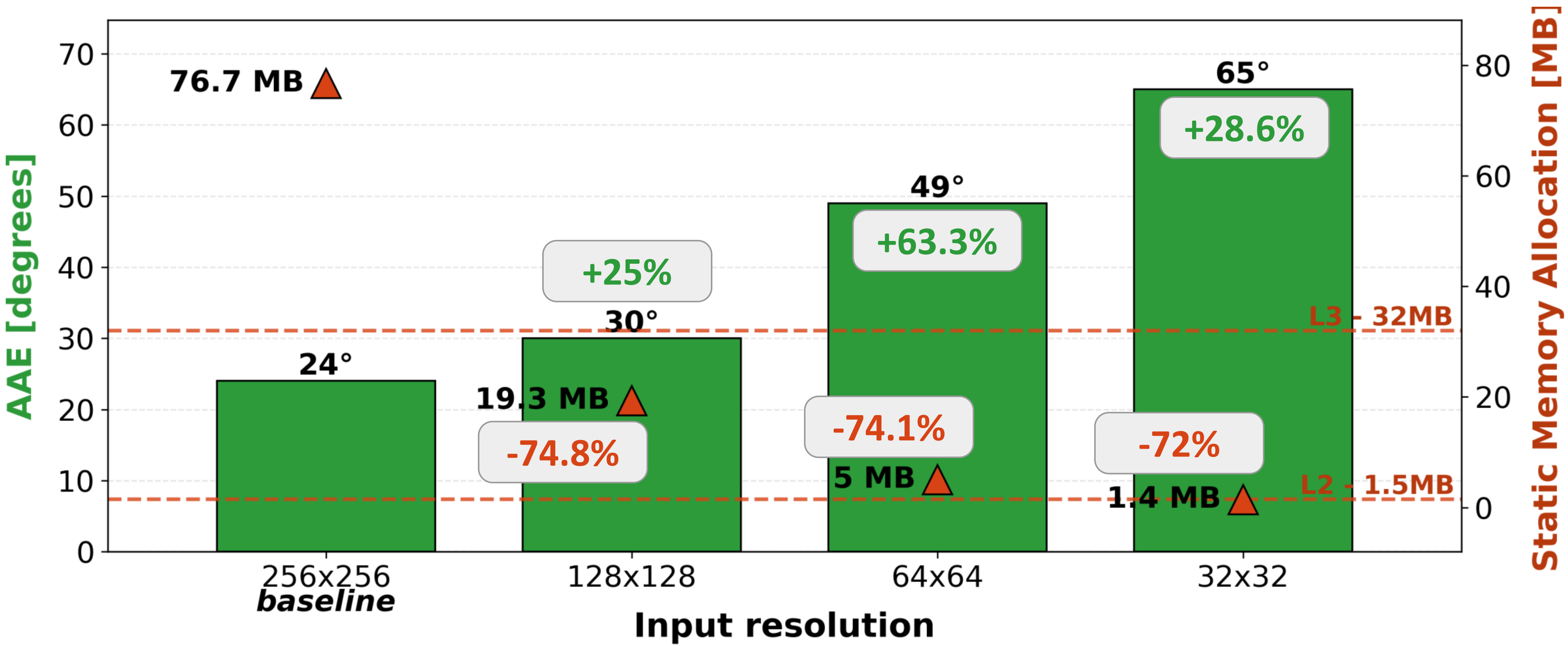
Results

Analysis on: INPUT RESOLUTION

ETH zürich
ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

Results

Analysis on: INPUT RESOLUTION



Results

Analysis on: QUANTIZATION (fp32 → int8)

uniform, per-tensor **Post-Training Quantization** on:
convolutions (activations, weights) and
LIF layers (parameters, activations and states)

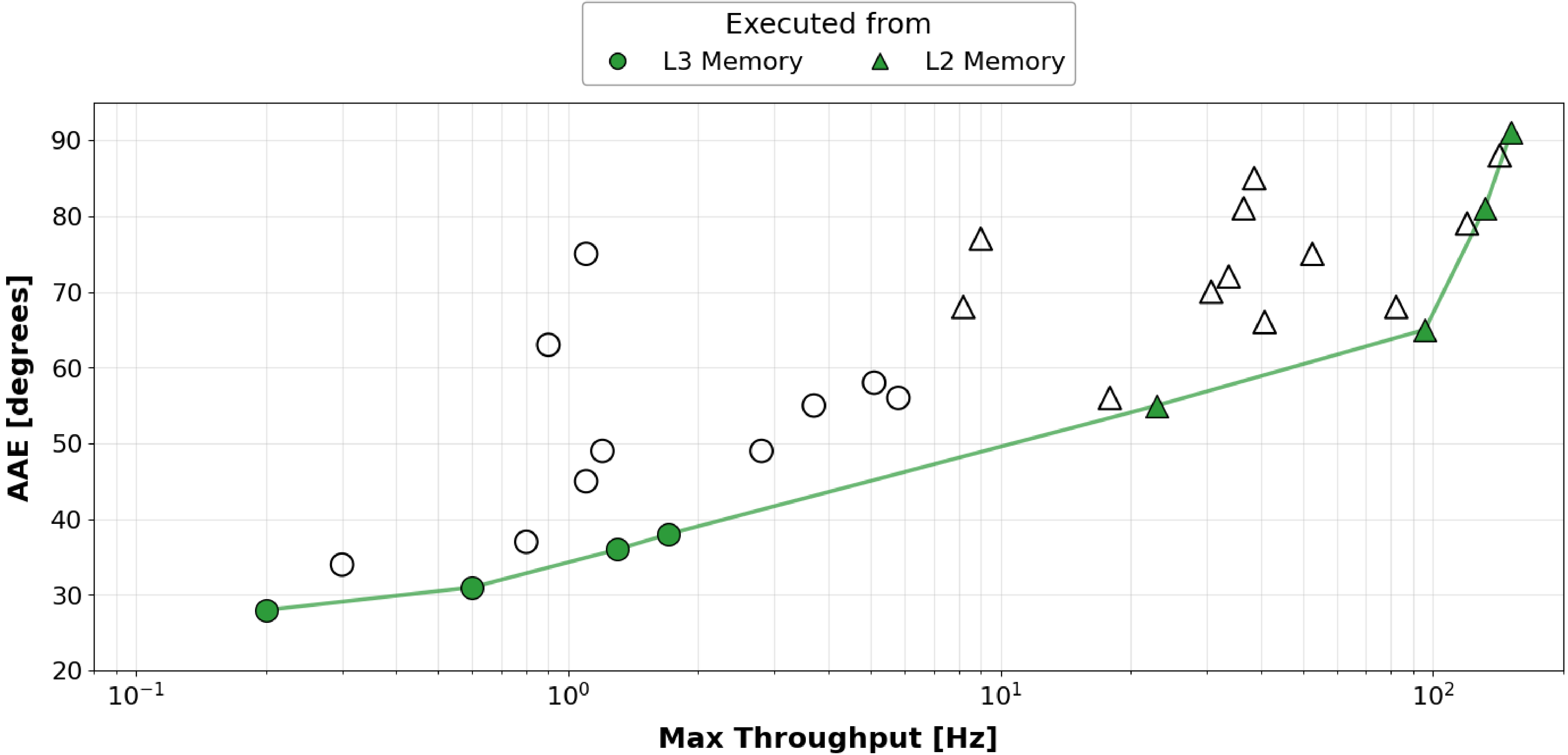
avg. memory allocation reduction	avg. error increase	avg. throughput increase
75%	**0.2%**	**x10^3**

Results Throughput vs. Error analysis [int8]

ETH zürich ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

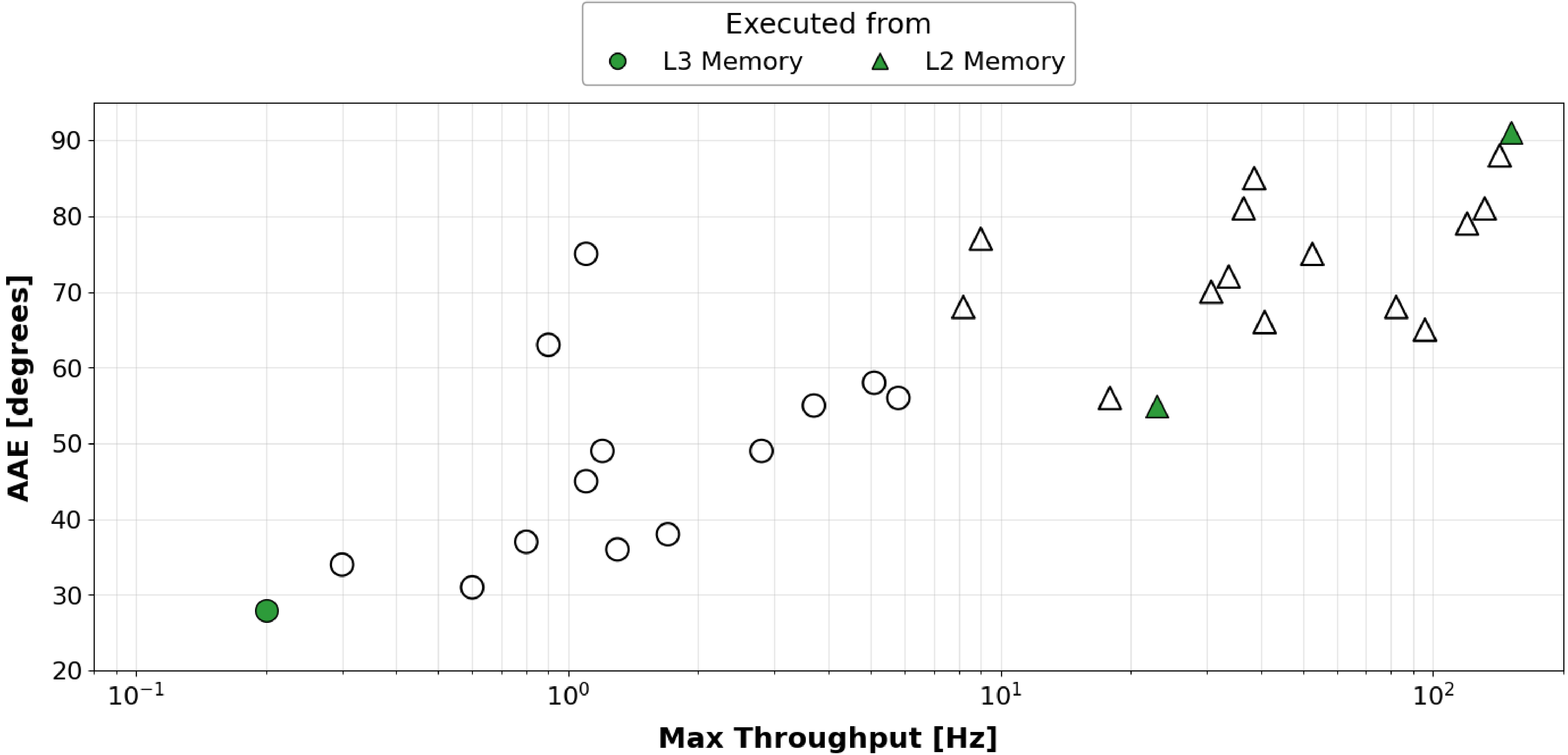
** throughput can be further improved by implementing L3-L2 double buffering with Deeploy*

Results Throughput vs. Error analysis [int8]: pareto curve



** throughput can be further improved by implementing L3-L2 double buffering with Deeploy*

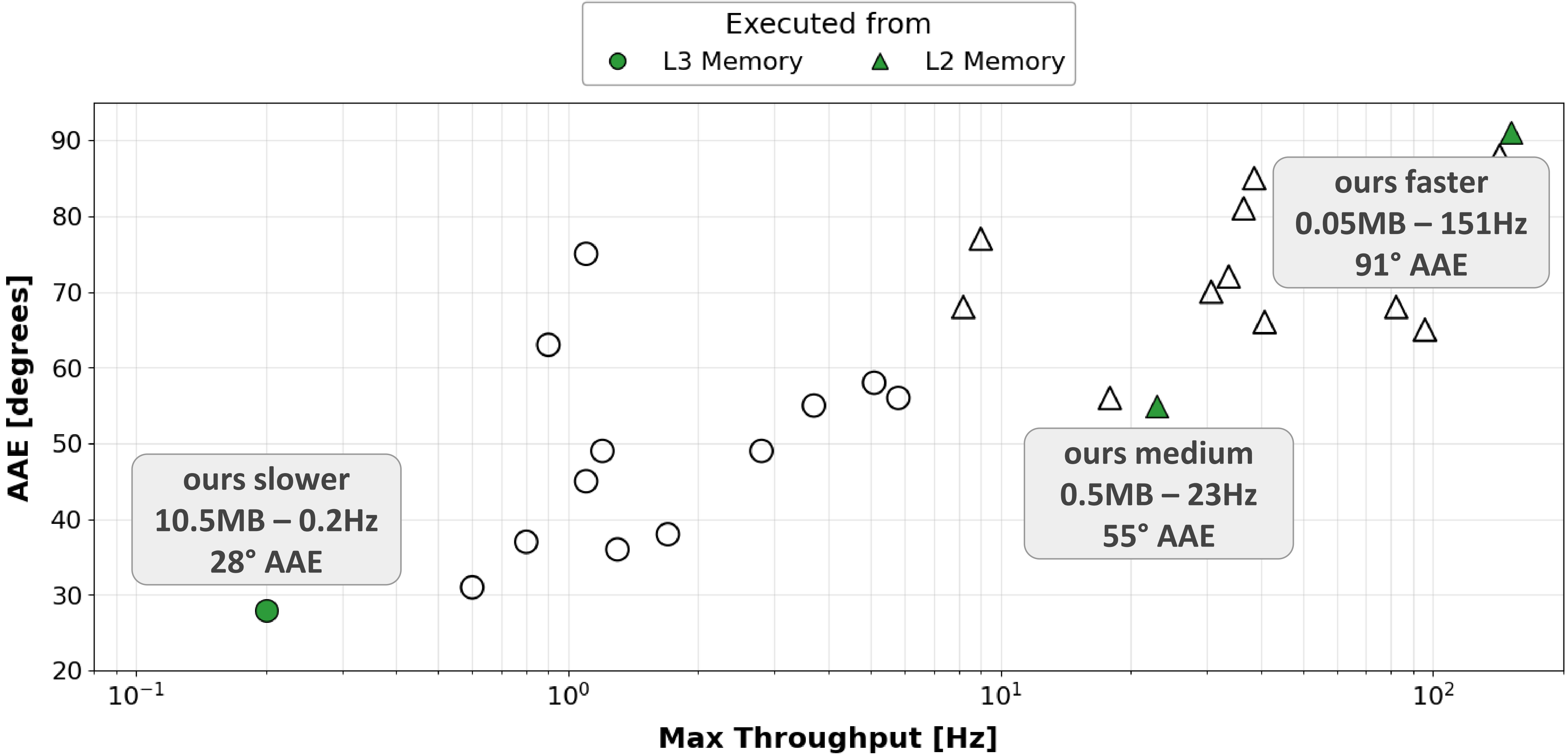
Results Throughput vs. Error analysis [int8]: pareto curve



ETH zürich ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA

** throughput can be further improved by implementing L3-L2 double buffering with Deeploy*

Results Throughput vs. Error analysis [int8]: pareto curve



* *throughput can be further improved by implementing L3-L2 double buffering with Deeploy*

Results

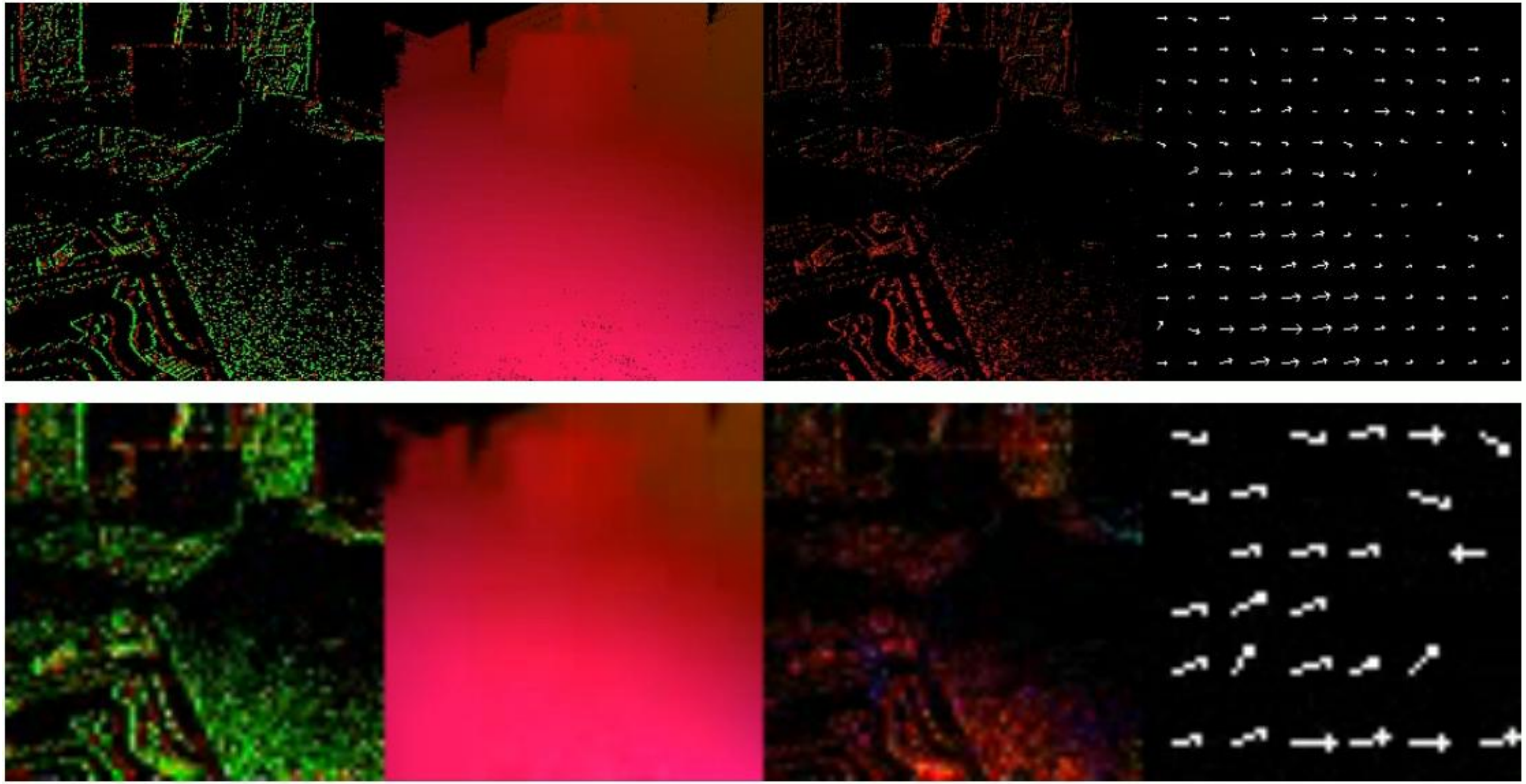
Model	Data type	AAE ↓ [degrees]	Static Memory Allocation [MB]	Max Throughput [Hz]
baseline	fp32	24°	76.7	0.025 *
ours medium	int8	**55°**	**0.5** ÷150	**23** ~ $x10^3$

* estimated (not deployable)

Results Optical flow in action:

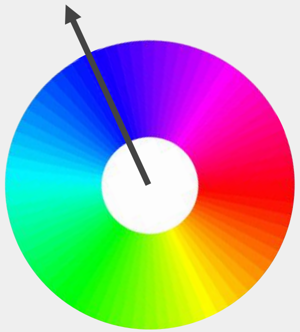
INPUT EVENTS GROUND-TRUTH FLOW ESTIMATED FLOW

baseline model, 256x256 pixels



our medium model, 64x64 pixels

LEGEND:
color = vector orientation



Conclusions

we employed **event-based cameras and spiking neural networks for optical flow estimation**, our contributions are:

SNN shrinking, from 80MB to 0.5MB (100x less)

Post training SNN **quantization** (conv, LIFs): accuracy loss <0.2%

Deployment on a ULP MCU: GAP9,
throughput from 0.1 to 150 FPS (100x higher)

to our knowledge, this is the first deployment of an event-based SNN for optical flow estimation on a general purpose ULP MCU (no ASIC).

Spiking optical flow is deployable

Q&A

Thank you for the attention!

ETH zürich

ALMA MATER STUDIORUM UNIVERSITÀ DI BOLOGNA